KronSpice architecture

A SPICE-compatible analog circuit simulator in pure Go, built as the fitness-evaluation engine for genetically evolved circuits

imports (main paths) · data flow · entry points · analysis / Kron · physics · GA layer · simulation core · leaf packages

cmd/kronspice
CLI: simulate a netlist → report / CSV
-solver · -all-channels · -o

cmd/evolve
CLI: run a GA from a JSON config
writes runs/<timestamp>-<name>/

examples/
amplifier_gain · filter_topology
torn_ga · kron_machine · 30 more

cmd/cir2kicad
CLI: netlist → KiCad 8 schematic
plus a results sheet

kicad/ · report/ · internal/export — schematics, results sheets
kicad imports netlist · report imports analysis, kron, circuit, netlist
export writes the netlist, schematic and results sheet for each example

evo/ — genetic algorithm layer
genome · encode_values (E-series) · encode_topology (parts grown into an embryo) · operators · fitness · checkpoint
Worker pool of runtime.NumCPU() · deterministic seeds · torn AC through kron · run directories, resume
Failed circuits (singular, no convergence, floating node, timeout) → penalty · weighted sum or NSGA-II

spacetime/ — gravity and fast rotation
gravity: Schwarzschild exterior, clock rates, signal redshift
optics: light delay and bending, effective medium n(r) for em/ field solvers
rotation: rotating frames and the fields a circuit sees
machine: one rotor under rotation and gravity
Keeps gravitational curvature apart from Kron's machine geometry
Leaf: imports nothing from the engine, callers wire its results in

analysis/ — analyses on the MNA system
op · dc sweep · ac (complex128 per frequency) · tran (adaptive step, LTE, breakpoints)
integrate: trapezoidal (default) · backward Euler
newton: limiting · reltol 1e-3 · itl1 100 / itl4 10 → gmin, source stepping, pseudo-transient
measure: rise time, gain, −3 dB, overshoot, settling, crossings
power: IEEE 1459, geometric-algebra and quaternion (→ hyper) · energy ledger (→ em)

kron/ — Kron tensors & diakoptics
primitive Z / Y · connection C · $Z' = C^T Z C$
mesh & node forms, twin-tested vs. MNA (1e-9)
orthogonal · exterior open paths · TAN chords
diakoptics: tear, cache, re-solve only the mutant
nonlinear · machine · gaframe · polyphase (→ hyper)

em/ — Maxwell's equations in quaternion form
nabla (Hamilton ∇) · maxwell (Treatise form) · biquat $\nabla F = J$ · vcalc twin
extract: lumped R, L, C, K · tline · inductance · biotsavart · coil · dipole
limits: electrical size against the lumped model · dispersion · multipactor
energy: open-system ledger · efficiency, Tellegen balance
lorentz ⟨F Q†⟩ · momentum · frames
duality · podolsky: hypothetical physics, labeled as such
analysis/ uses the ledger; field code does not run inside Newton / LU

Result → checks

solver/ — linear solve (leaf: imports nothing)
dense LU, real and complex, partial pivoting: below 30 unknowns
sparse Gilbert–Peierls, minimum-degree ordering: 30 and above, or forced with Options.Backend
Imported by analysis, kron and evo. Newton iteration lives in analysis/

output/ — results
result: in-memory vectors + energy ledger
text reports · CSV
one Result type shared by MNA, kron and em

hyper/ — hypercomplex numbers (stdlib only)
quat: Hamilton quaternions, products, rotors, matrices, exp / log
ga: Clifford algebra Cl(p,q,r), bitmask blades · G2, G3 + sparse multivector
Never imported by solver or device: inner loop stays float64 / complex128

device/ — one file per device family, stamping through Stamper
R C L K · V I sources (DC PULSE SIN PWL EXP AC) · controlled E F G H · nonlinear B · gyrator Y · machine N · bearing · charge- and flux-defined C, L
diode (SPICE3 level 1) · bjt (Gummel–Poon) · mosfet levels 1–3
Device: Setup · Load · LoadAC, with optional Reactive and Nonlinear — every device embeds device.Base, so it carries the full external port set

fluid/ (leaf)
media: H_2O and D_2O properties
acoustic: piston radiation,
flask breathing mode

material/ (leaf)
cited constants: Seebeck,
resistivities, electrode potentials,
emission lines

circuit/ — in-memory circuit model
nodes, elements, models · builder.go: programmatic API used by evo
built from netlist/ or the Builder · Validate() before every simulation

netlist/ — SPICE text input
parser · .op .dc .ac .tran .model .title .domain
units.go: engineering suffixes, meg, mil

lumped values (em) · media and loads (fluid) → wired into circuit.Builder by callers

exchange/ — external energy ports on every element (leaf, stdlib only · imported by device, circuit, output, em, analysis and kron)

Every element exposes one ExternalPort per registered channel. No model means exactly 0 W, listed as NotModeled. In Kron's terms each port is an open path to the exterior junction Ω of its channel.

Electrical	Light	Thermal	Magnetic field	Electric field	Mechanical	Acoustic	Ionizing radiation	Chemical	Neutrino	Gravity / frame	Magnetic charge	Future 1 … 4
modeled (MNA)	in one example	device heat	not modeled yet	not modeled yet	domain, machines	domain	not modeled yet	in one example	not modeled yet	no energy	hypothetical only	reserved

Evaluating one candidate

Genome → circuit.Builder → Validate() → Setup → Newton + LU → output.Result → ledger + measure → fitness

every evaluation runs under a context.Context deadline · AC can be solved torn, with pieces cached between candidates
any error → typed error → penalty fitness, so the GA learns from broken circuits

Open-system energy ledger

$dW_{stored}/dt = P_{in} - P_{out} - P_{dissipated}$

Kept per element × channel
Residual above tolerance ⇒ reported as an error
Model-supplied (E/F/G/H) and hypothetical power in their own columns
Efficiency (operating point and transient) · Tellegen's sum
Evolution cannot profit from energy appearing from nowhere
Twin check: Kron open-path ledger = MNA ledger, row by row

Design rules

Never panic or hang: typed errors, context deadlines
Deterministic: same netlist, options and seed, identical results
Parallel: no shared mutable state, passes the race detector
Electrical-size check for the lumped limit of Maxwell's equations
Solver core float64 / complex128, allocation-free solve
Golden tests vs. ngspice; twin tests for Kron, GA, quaternions
Pure Go, no cgo, no third-party dependencies

Not built yet (design in CLAUDE.md): env/ outside influences · plasma/ and arc models · bubble dynamics · FDTD and TLM field solvers · .subckt, .param, .include · switches · transmission-line devices · SPICE3 raw output · QSpice cross-check